Scanning and Enumeration Lab Report

Target Environment: Kali Linux Attack VM vs. Two Vulnerable Targets (192.168.255.0/24)

Course: CIS4204 Ethical Hacking

Student: Kyle Potchka

Date: September 26, 2026

Introduction

In this lab I used Nmap to run a series of ping sweeps and port scans described in Chapter 3, against two vulnerable hosts in my own private virtual lab environment, built in an earlier lab using VirtualBox. Each of the eight questions below explains what a specific Nmap switch does, followed by the exact command(s) I ran to observe that switch in action and a screen capture of the actual output. No scans in this lab were directed at any host outside my own private lab network.

The lab environment consists of three VirtualBox VMs on a shared network (bridged adapter, subnet 192.168.255.0/24, gateway 192.168.255.1):

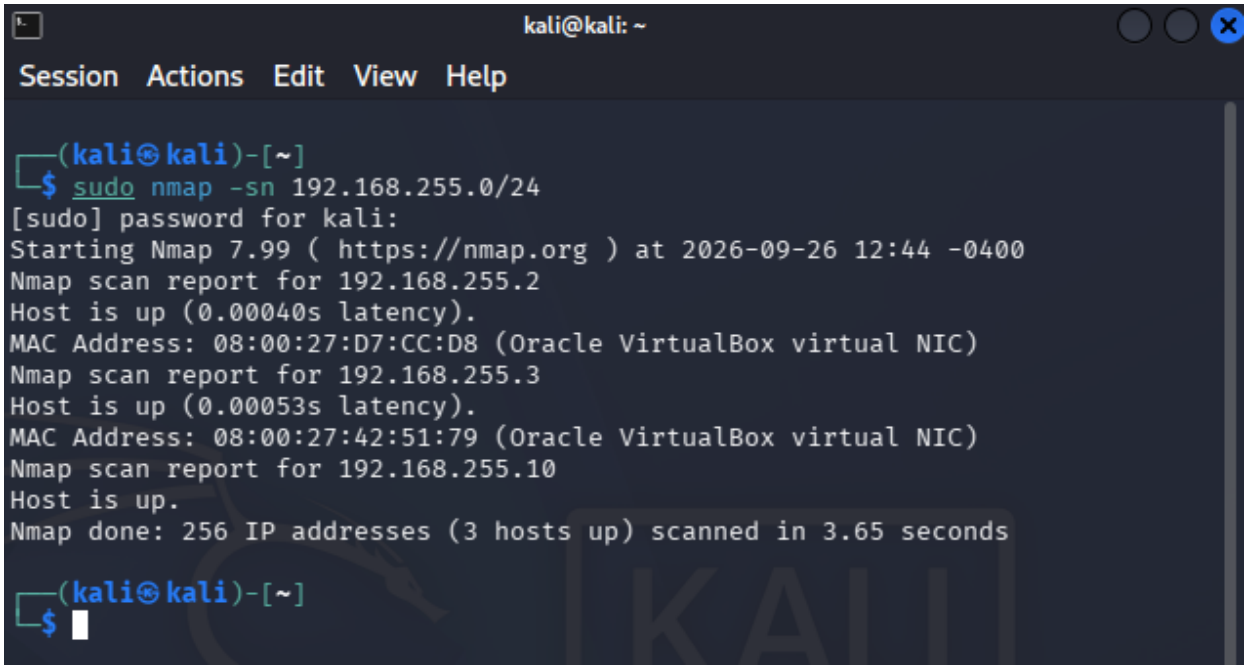
- Kali Linux (attack machine) — 192.168.255.10
- Metasploitable2-windows2008 (vulnerable Windows Server 2008 target) — 192.168.255.2
- Metasploitable3-ub1404 (vulnerable Ubuntu Linux target) — 192.168.255.3

1. Explain what the -sn switch does.

The -sn switch tells Nmap to perform host discovery only, with no port scan. Nmap determines which hosts on the specified range are online and responding, without probing any ports on them. By default it does this using a combination of an ICMP echo request, a TCP SYN packet to port 443, a TCP ACK packet to port 80, and an ICMP timestamp request; if the target answers any of these, it is reported as "up." This switch was formerly called -sP. It is the fastest way to build a live-host inventory of a subnet before running more detailed scans against individual hosts.

Scan 1 of 17 — I ran this from a terminal on my Kali machine:

```
sudo nmap -sn 192.168.255.0/24
```



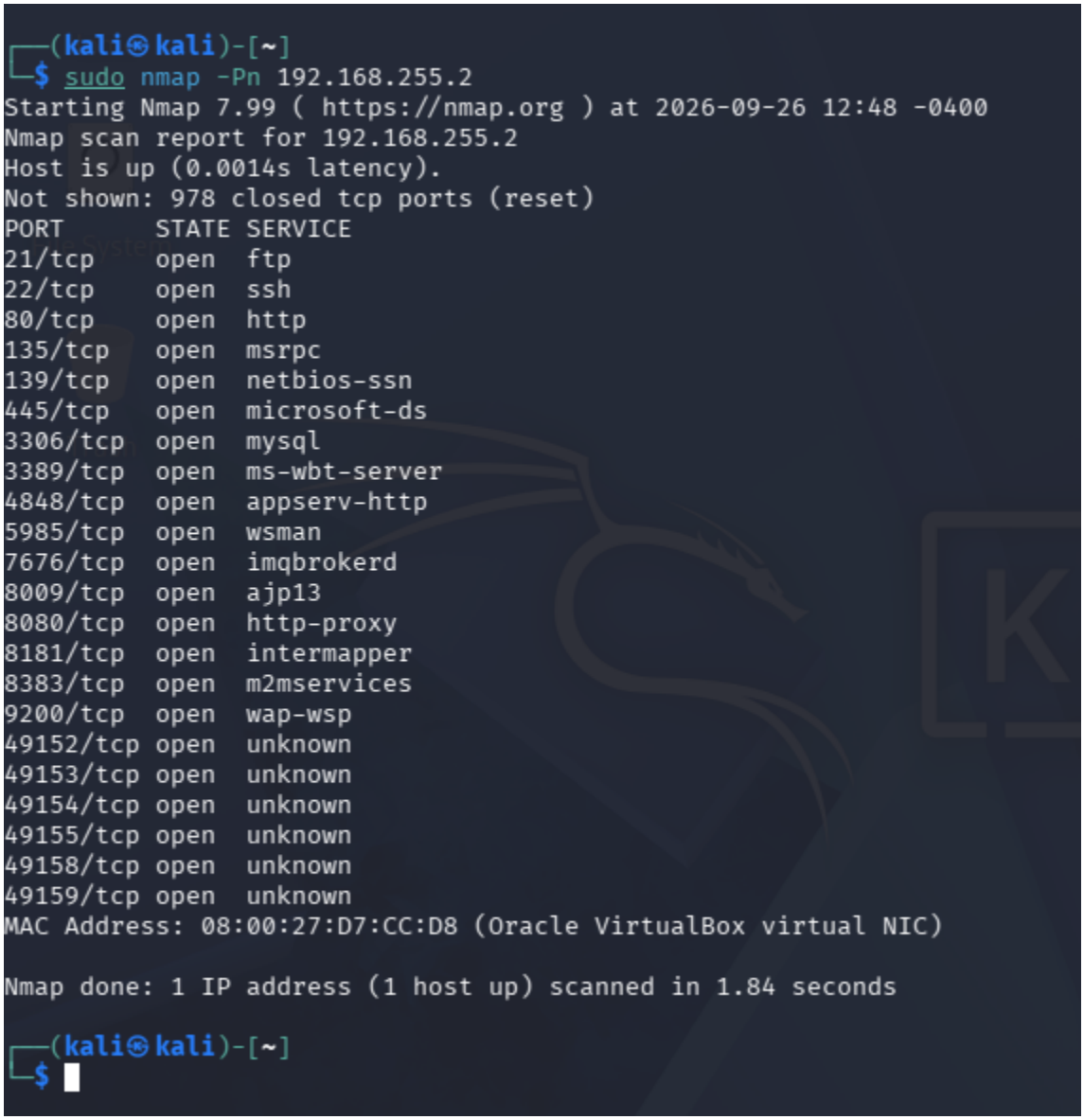
Ping-sweep of the whole lab subnet — screenshot should show every host reported "up," including Kali and Metasploitable.

2. Explain what the -Pn switch does.

The -Pn switch skips the host-discovery (ping) phase entirely and tells Nmap to treat every target IP as if it were online, moving straight into port scanning. This is useful against hosts or firewalls configured to drop ICMP echo requests and the other discovery probes Nmap normally relies on — without -Pn, such a host could be incorrectly reported as down and skipped, even though its ports are reachable and scannable.

Scan 2 of 17 — I ran this from a terminal on my Kali machine:

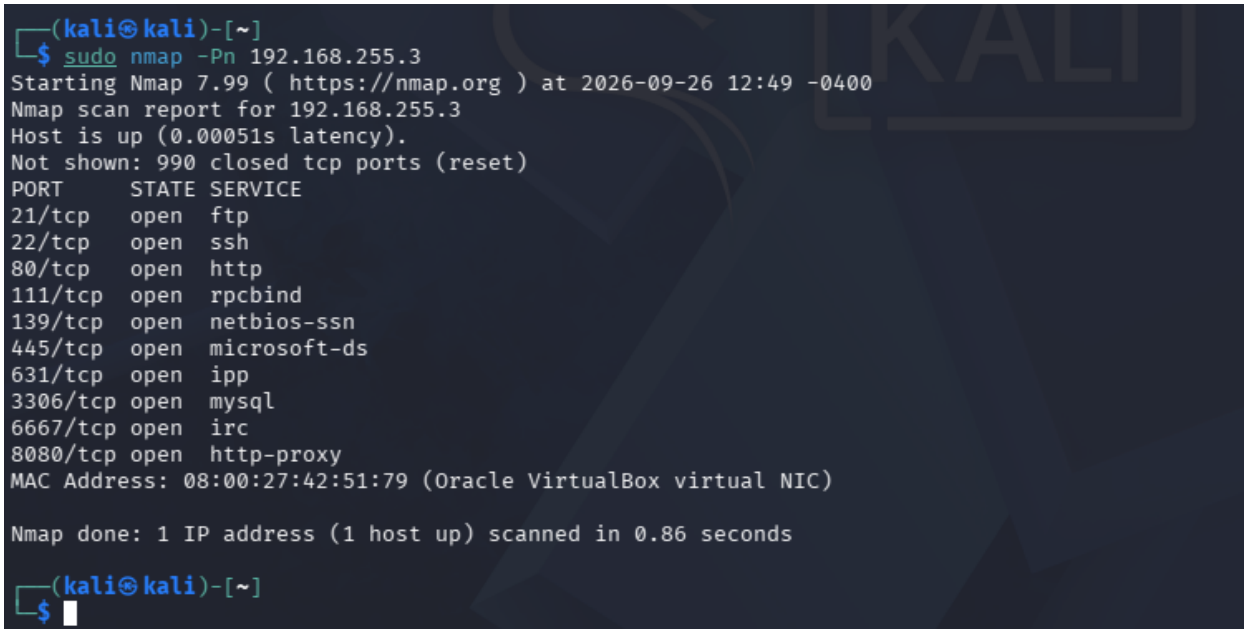
```
sudo nmap -Pn 192.168.255.2
```



Skips host discovery and scans the default port list against .2.

Scan 3 of 17 — I ran this from a terminal on my Kali machine:

```
sudo nmap -Pn 192.168.255.3
```



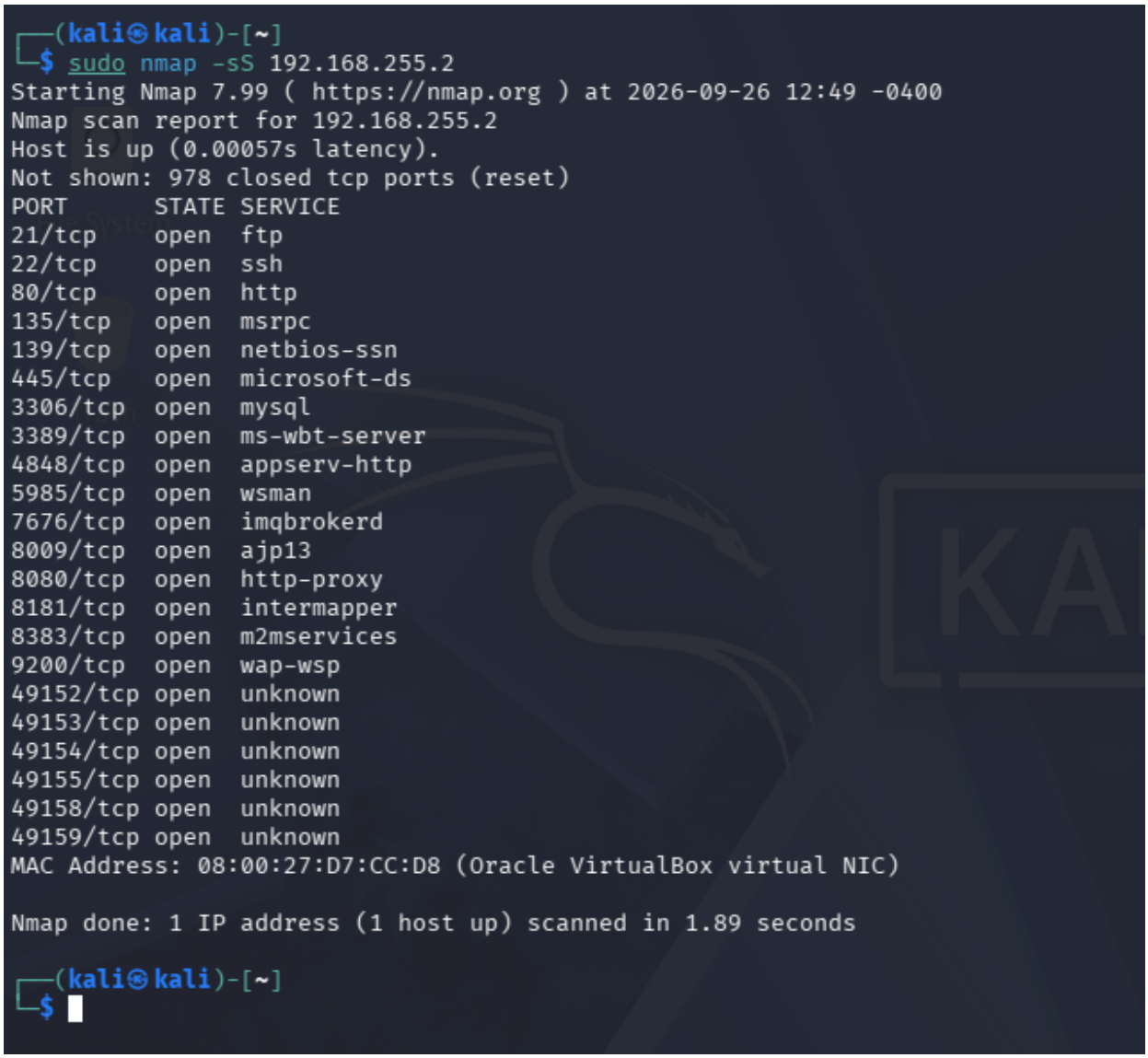
Same, run against .3.

3. Explain what the -sS switch does.

The -sS switch runs a TCP SYN scan, also called a half-open or “stealth” scan. For each target port, Nmap sends a single SYN packet; a SYN/ACK response means the port is open (Nmap then sends a RST instead of completing the handshake with an ACK), a RST response means the port is closed, and no response or an ICMP error means the port is filtered. Because the full TCP three-way handshake is never completed, the scan is faster and less likely to be logged by the target application than a full connect scan. This is Nmap's default scan type when run with sufficient (root/raw-socket) privileges, which is why the commands are run with sudo.

Scan 4 of 17 — I ran this from a terminal on my Kali machine:

```
sudo nmap -sS 192.168.255.2
```



SYN scan against .2 — shows open/closed/filtered ports found via half-open connections.

Scan 5 of 17 — I ran this from a terminal on my Kali machine:

```
sudo nmap -sS 192.168.255.3
```

```
┌──(kali㉿kali)-[~]
└─$ sudo nmap -sS 192.168.255.3
Starting Nmap 7.99 ( https://nmap.org ) at 2026-09-26 12:51 -0400
Nmap scan report for 192.168.255.3
Host is up (0.00046s latency).
Not shown: 990 closed tcp ports (reset)
PORT     STATE SERVICE
21/tcp   open  ftp
22/tcp   open  ssh
80/tcp   open  http
111/tcp  open  rpcbind
139/tcp  open  netbios-ssn
445/tcp  open  microsoft-ds
631/tcp  open  ipp
3306/tcp open  mysql
6667/tcp open  irc
8080/tcp open  http-proxy
MAC Address: 08:00:27:42:51:79 (Oracle VirtualBox virtual NIC)

Nmap done: 1 IP address (1 host up) scanned in 1.04 seconds
```

SYN scan against .3.

4. Explain what the -sT and -sU switches do.

-sT runs a TCP Connect scan, which completes the full TCP three-way handshake (SYN, SYN/ACK, ACK) for every target port using the operating system's normal connect() call. It doesn't require raw-socket privileges the way -sS does, but it is slower and far more likely to be logged, since it opens a complete connection to each port. -sU runs a UDP scan instead of a TCP scan: Nmap sends UDP packets to each target port and interprets the response — an ICMP “port unreachable” message means the port is closed, a UDP response means it is open, and no response at all is reported as open|filtered, since UDP does not acknowledge packets the way TCP does. UDP scanning matters because -sS and -sT only examine TCP ports, and many important services (DNS, SNMP, DHCP, TFTP, etc.) run over UDP.

Scan 6 of 17 — I ran this from a terminal on my Kali machine:

```
sudo nmap -sT 192.168.255.2
```

```
┌──(kali㉿kali)-[~]
└─$ sudo nmap -sT 192.168.255.2
Starting Nmap 7.99 ( https://nmap.org ) at 2026-09-26 12:52 -0400
Nmap scan report for 192.168.255.2
Host is up (0.0027s latency).
Not shown: 978 closed tcp ports (conn-refused)
PORT      STATE SERVICE
21/tcp    open  ftp
22/tcp    open  ssh
80/tcp    open  http
135/tcp   open  msrpc
139/tcp   open  netbios-ssn
445/tcp   open  microsoft-ds
3306/tcp  open  mysql
3389/tcp  open  ms-wbt-server
4848/tcp  open  appserv-http
5985/tcp  open  wsman
7676/tcp  open  imqbrokerd
8009/tcp  open  ajp13
8080/tcp  open  http-proxy
8181/tcp  open  intermapper
8383/tcp  open  m2mservices
9200/tcp  open  wap-wsp
49152/tcp open  unknown
49153/tcp open  unknown
49154/tcp open  unknown
49155/tcp open  unknown
49158/tcp open  unknown
49159/tcp open  unknown
MAC Address: 08:00:27:D7:CC:D8 (Oracle VirtualBox virtual NIC)

Nmap done: 1 IP address (1 host up) scanned in 1.78 seconds
```

Full TCP connect scan against .2 — compare open ports to the -sS results.

Scan 7 of 17 — I ran this from a terminal on my Kali machine:

```
sudo nmap -sU 192.168.255.2
```

```
┌──(kali㉿kali)-[~]
└─$ sudo nmap -sU 192.168.255.2
Starting Nmap 7.99 ( https://nmap.org ) at 2026-09-26 12:54 -0400
Stats: 0:01:50 elapsed; 0 hosts completed (1 up), 1 undergoing UDP Scan
UDP Scan Timing: About 49.60% done; ETC: 12:58 (0:01:52 remaining)
Stats: 0:01:51 elapsed; 0 hosts completed (1 up), 1 undergoing UDP Scan
UDP Scan Timing: About 49.80% done; ETC: 12:58 (0:01:53 remaining)
Stats: 0:03:45 elapsed; 0 hosts completed (1 up), 1 undergoing UDP Scan
UDP Scan Timing: About 58.90% done; ETC: 13:01 (0:02:38 remaining)
Stats: 0:06:20 elapsed; 0 hosts completed (1 up), 1 undergoing UDP Scan
UDP Scan Timing: About 72.50% done; ETC: 13:03 (0:02:24 remaining)
Stats: 0:07:07 elapsed; 0 hosts completed (1 up), 1 undergoing UDP Scan
UDP Scan Timing: About 76.80% done; ETC: 13:04 (0:02:09 remaining)
Stats: 0:10:18 elapsed; 0 hosts completed (1 up), 1 undergoing UDP Scan
UDP Scan Timing: About 94.10% done; ETC: 13:05 (0:00:39 remaining)
Nmap scan report for 192.168.255.2
Host is up (0.00088s latency).
Not shown: 992 closed udp ports (port-unreach)
PORT     STATE         SERVICE
123/udp  open|filtered ntp
137/udp  open          netbios-ns
138/udp  open|filtered netbios-dgm
161/udp  open          snmp
500/udp  open|filtered isakmp
4500/udp open|filtered nat-t-ike
5353/udp open|filtered zeroconf
5355/udp open|filtered llmnr
MAC Address: 08:00:27:D7:CC:D8 (Oracle VirtualBox virtual NIC)

Nmap done: 1 IP address (1 host up) scanned in 708.67 seconds

┌──(kali㉿kali)-[~]
└─$
```

UDP scan against .2 — typically slower; look for open or open|filtered UDP ports.

Scan 8 of 17 — I ran this from a terminal on my Kali machine:

```
sudo nmap -sT 192.168.255.3
```

```
┌──(kali㉿kali)-[~]
└─$ sudo nmap -sT 192.168.255.3
Starting Nmap 7.99 ( https://nmap.org ) at 2026-09-26 13:09 -0400
Nmap scan report for 192.168.255.3
Host is up (0.0028s latency).
Not shown: 990 closed tcp ports (conn-refused)
PORT     STATE SERVICE
21/tcp   open  ftp
22/tcp   open  ssh
80/tcp   open  http
111/tcp  open  rpcbind
139/tcp  open  netbios-ssn
445/tcp  open  microsoft-ds
631/tcp  open  ipp
3306/tcp open  mysql
6667/tcp open  irc
8080/tcp open  http-proxy
MAC Address: 08:00:27:42:51:79 (Oracle VirtualBox virtual NIC)

Nmap done: 1 IP address (1 host up) scanned in 1.01 seconds
```

Full TCP connect scan against .3.

Scan 9 of 17 — I ran this from a terminal on my Kali machine:

```
sudo nmap -sU 192.168.255.3
```

```
┌──(kali㉿kali)-[~]
└─$ sudo nmap -sU 192.168.255.3
Starting Nmap 7.99 ( https://nmap.org ) at 2026-09-26 13:09 -0400
Stats: 0:06:54 elapsed; 0 hosts completed (1 up), 1 undergoing UDP Scan
UDP Scan Timing: About 41.17% done; ETC: 13:26 (0:09:52 remaining)
Stats: 0:17:11 elapsed; 0 hosts completed (1 up), 1 undergoing UDP Scan
UDP Scan Timing: About 97.77% done; ETC: 13:27 (0:00:24 remaining)
Nmap scan report for 192.168.255.3
Host is up (0.00093s latency).
Not shown: 995 closed udp ports (port-unreach)
PORT     STATE         SERVICE
111/udp  open          rpcbind
137/udp  open          netbios-ns
138/udp  open|filtered netbios-dgm
631/udp  open|filtered ipp
5353/udp open          zeroconf
MAC Address: 08:00:27:42:51:79 (Oracle VirtualBox virtual NIC)

Nmap done: 1 IP address (1 host up) scanned in 1080.01 seconds

┌──(kali㉿kali)-[~]
└─$
```

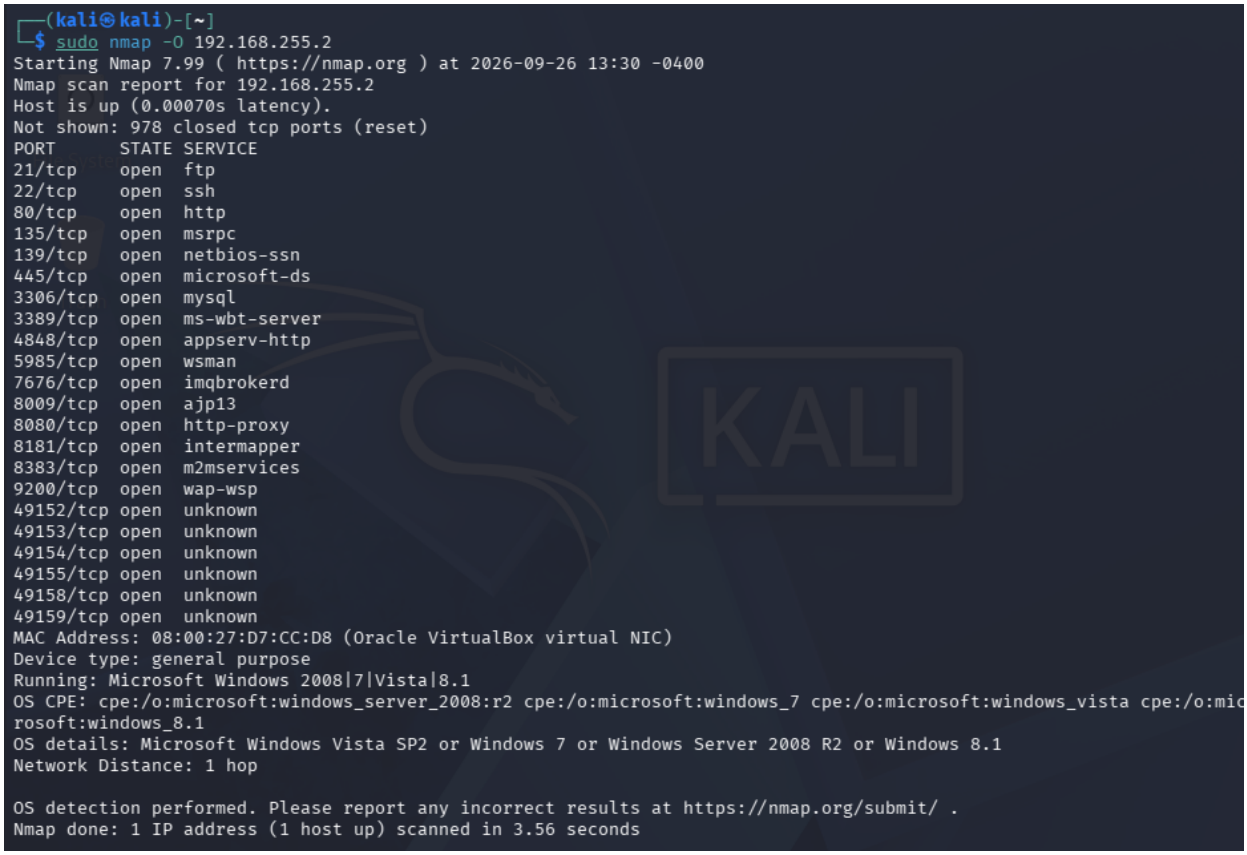
UDP scan against .3.

5. Explain what the -O switch does.

The -O switch enables Nmap's OS detection engine. Nmap sends a series of specially crafted TCP, UDP, and ICMP probes to the target and examines subtle characteristics of the replies — initial TTL values, TCP window sizes, the order and options in the TCP header, ISN sampling, and more — to build a fingerprint of the target's TCP/IP stack. It then compares that fingerprint against its database of known OS fingerprints and reports its best guesses, often with a confidence percentage, for the operating system and sometimes the device type running on the target.

Scan 10 of 17 — I ran this from a terminal on my Kali machine:

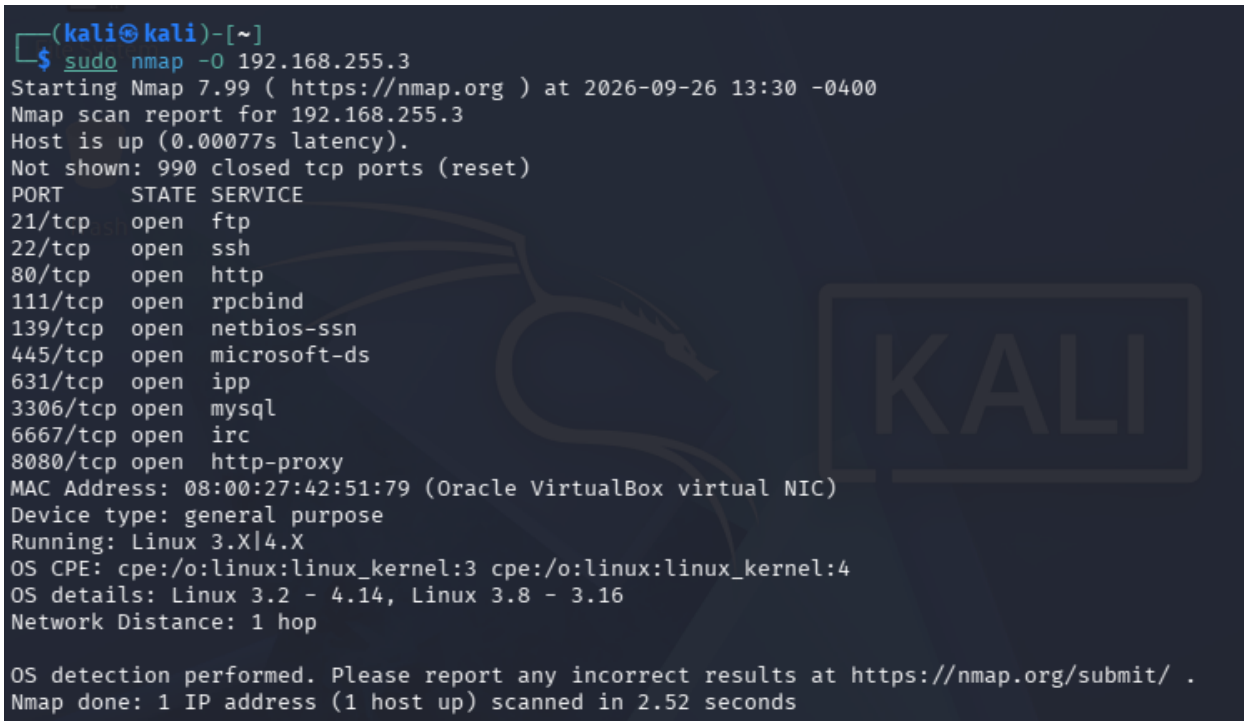
```
sudo nmap -O 192.168.255.2
```



OS fingerprinting against .2 — should show Nmap's OS guess(es) and confidence.

Scan 11 of 17 — I ran this from a terminal on my Kali machine:

```
sudo nmap -O 192.168.255.3
```



OS fingerprinting against .3.

6. Explain what the -sV switch does.

The -sV switch enables version detection. After finding open ports, Nmap goes a step further than reporting just “port 22 open” — it interacts with each open port using probes from its service-probe database to determine the actual application and version listening there, for example OpenSSH 7.6p1 on port 22 or Apache httpd 2.4.29 on port 80. This is far more useful for vulnerability assessment than the port number alone, since known vulnerabilities are usually tied to specific software versions.

Scan 12 of 17 — I ran this from a terminal on my Kali machine:

```
sudo nmap -sV 192.168.255.2
```

```
┌──(kali㉿kali)-[~]
└─$ sudo nmap -sV 192.168.255.2
Starting Nmap 7.99 ( https://nmap.org ) at 2026-09-26 13:31 -0400
Nmap scan report for 192.168.255.2
Host is up (0.0016s latency).
Not shown: 978 closed tcp ports (reset)
PORT      STATE SERVICE             VERSION
21/tcp    open  ftp                 Microsoft ftpd
22/tcp    open  ssh                 OpenSSH 7.1 (protocol 2.0)
80/tcp    open  http                Microsoft IIS httpd 7.5
135/tcp   open  msrpc               Microsoft Windows RPC
139/tcp   open  netbios-ssn         Microsoft Windows netbios-ssn
445/tcp   open  microsoft-ds        Microsoft Windows Server 2008 R2 - 2012 microsoft-ds
3306/tcp  open  mysql               MySQL 5.5.20-log
3389/tcp  open  ms-wbt-server?
4848/tcp  open  ssl/http            Oracle Glassfish Application Server
5985/tcp  open  http                Microsoft HTTPAPI httpd 2.0 (SSDP/UPnP)
7676/tcp  open  java-message-service Java Message Service 301
8009/tcp  open  ajp13               Apache Jserv (Protocol v1.3)
8080/tcp  open  http                Sun GlassFish Open Source Edition  4.0
8181/tcp  open  ssl/intermapper?
8383/tcp  open  http                Apache httpd
9200/tcp  open  http                Elasticsearch REST API 1.1.1 (name: Living Brain; Lucene 4.7)
49152/tcp open  msrpc               Microsoft Windows RPC
49153/tcp open  msrpc               Microsoft Windows RPC
49154/tcp open  msrpc               Microsoft Windows RPC
49155/tcp open  msrpc               Microsoft Windows RPC
49158/tcp open  java-rmi            Java RMI
49159/tcp open  tcpwrapped
MAC Address: 08:00:27:D7:CC:D8 (Oracle VirtualBox virtual NIC)
Service Info: OSs: Windows, Windows Server 2008 R2 - 2012; CPE: cpe:/o:microsoft:windows

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 86.64 seconds

┌──(kali㉿kali)-[~]
```

Version detection against .2 — should show service names/versions next to each open port.

Scan 13 of 17 — I ran this from a terminal on my Kali machine:

```
sudo nmap -sV 192.168.255.3
```

```
┌──(kali㉿kali)-[~]
└─$ sudo nmap -sV 192.168.255.3
Starting Nmap 7.99 ( https://nmap.org ) at 2026-09-26 13:34 -0400
Nmap scan report for 192.168.255.3
Host is up (0.00041s latency).
Not shown: 990 closed tcp ports (reset)
PORT     STATE SERVICE     VERSION
21/tcp   open  ftp         ProFTPD 1.3.5
22/tcp   open  ssh         OpenSSH 6.6.1p1 Ubuntu 2ubuntu2.13 (Ubuntu Linux; protocol 2.0)
80/tcp   open  http        Apache httpd 2.4.7
111/tcp  open  rpcbind     2-4 (RPC #100000)
139/tcp  open  netbios-ssn Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
445/tcp  open  netbios-ssn Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
631/tcp  open  ipp         CUPS 1.7
3306/tcp open  mysql       MySQL (unauthorized)
6667/tcp open  irc         UnrealIRCd
8080/tcp open  http        Jetty 8.1.7.v20120910
MAC Address: 08:00:27:42:51:79 (Oracle VirtualBox virtual NIC)
Service Info: Hosts: 127.0.0.1, METASPLOITABLE3-UB1404, irc.TestIRC.net; OSs: Unix, Linux; CPE: cpe:/o:linux:linux_k
ernel

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 12.72 seconds
```

Version detection against .3.

7. Explain what the -A switch does.

The -A switch is a convenience/aggressive-scan option that turns on several intensive features at once: OS detection (equivalent to -O), service/version detection (equivalent to -sV), default Nmap Scripting Engine scripts (equivalent to -sC), and a traceroute to the target. It produces a much more complete picture of the target in a single command, but because it combines so many probes and scripts it is slower and considerably easier for an intrusion detection system to notice than a plain SYN scan.

Scan 14 of 17 — I ran this from a terminal on my Kali machine:

```
sudo nmap -A 192.168.255.2
```



Aggressive scan against .2 — combines OS + version detection, default scripts, and traceroute; this output will be the longest.

Scan 15 of 17 — I ran this from a terminal on my Kali machine:

```
sudo nmap -A 192.168.255.3
```

```
┌──(kali㉿kali)-[~]
└─$ sudo nmap -A 192.168.255.3
Starting Nmap 7.99 ( https://nmap.org ) at 2026-09-26 14:03 -0400
Nmap scan report for 192.168.255.3
Host is up (0.00088s latency).
Not shown: 990 closed tcp ports (reset)
PORT     STATE SERVICE     VERSION
21/tcp   open  ftp         ProFTPD 1.3.5
22/tcp   open  ssh         OpenSSH 6.6.1p1 Ubuntu 2ubuntu2.13 (Ubuntu Linux; protocol 2.0)
| ssh-hostkey:
|   1024 2b:2e:1f:a4:54:26:87:76:12:26:59:58:0d:da:3b:04 (DSA)
|   2048 c9:ac:70:ef:f8:de:8b:a3:a3:44:ab:3d:32:0a:5c:6a (RSA)
|   256 c0:49:cc:18:7b:27:a4:07:0d:2a:0d:bb:42:4c:36:17 (ECDSA)
|_  256 a0:76:f3:76:f8:f0:70:4d:09:ca:e1:10:fd:a9:cc:0a (ED25519)
80/tcp   open  http        Apache httpd 2.4.7
|_http-server-header: Apache/2.4.7 (Ubuntu)
|_http-title: Index of /
| http-ls: Volume /
| SIZE  TIME              FILENAME
| -     2020-10-29 19:37  chat/
| -     2011-07-27 20:17  drupal/
| 1.7K  2020-10-29 19:37  payroll_app.php
| -     2013-04-08 12:06  phpmyadmin/
|_
111/tcp  open  rpcbind     2-4 (RPC #100000)
| rpcinfo:
|   program version    port/proto  service
|   100000  2,3,4        111/tcp   rpcbind
|   100000  2,3,4        111/udp   rpcbind
|   100000  3,4          111/tcp6  rpcbind
|   100000  3,4          111/udp6  rpcbind
|   100024  1          38361/udp6  status
|   100024  1          42480/udp   status
|   100024  1          44980/tcp6  status
|_  100024  1          57757/tcp   status
139/tcp  open  netbios-ssn Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
445/tcp  open  netbios-ssn Samba smbd 4.3.11-Ubuntu (workgroup: WORKGROUP)
631/tcp  open  ipp         CUPS 1.7
| http-methods:
|_  Potentially risky methods: PUT
|_http-server-header: CUPS/1.7 IPP/2.1
|_http-title: Home - CUPS 1.7.2
| http-robots.txt: 1 disallowed entry
|_/
3306/tcp open  mysql       MySQL (unauthorized)
6667/tcp open  irc         UnrealIRCd
8080/tcp open  http        Jetty 8.1.7.v20120910
|_http-server-header: Jetty(8.1.7.v20120910)
|_http-title: Error 404 - Not Found
MAC Address: 08:00:27:42:51:79 (Oracle VirtualBox virtual NIC)
Device type: general purpose
Running: Linux 3.X|4.X
OS CPE: cpe:/o:linux:linux_kernel:3 cpe:/o:linux:linux_kernel:4
OS details: Linux 3.2 - 4.14
Network Distance: 1 hop
Service Info: Hosts: 127.0.0.1, METASPLOITABLE3-UB1404, irc.TestIRC.net; OSs: Unix, Linux; CPE: cpe:/o:linux:linux_kernel

Host script results:
| smb-os-discovery:
|   OS: Windows 6.1 (Samba 4.3.11-Ubuntu)
|   Computer name: metasploitable3-ub1404
|   NetBIOS computer name: METASPLOITABLE3-UB1404\x00
|   Domain name: \x00
|   FQDN: metasploitable3-ub1404
|_  System time: 2026-09-26T18:03:53+00:00
|_clock-skew: mean: -2s, deviation: 0s, median: -2s
| smb2-time:
|   date: 2026-09-26T18:03:53
|_  start_date: N/A
| smb2-security-mode:
|   3.1.1:
|_    Message signing enabled but not required
| smb-security-mode:
|   account_used: guest
|   authentication_level: user
|   challenge_response: supported
|_  message_signing: disabled (dangerous, but default)
|_nbstat: NetBIOS name: METASPLOITABLE3, NetBIOS user: <unknown>, NetBIOS MAC: <unknown> (unknown)

TRACEROUTE
HOP RTT     ADDRESS
1   0.88 ms 192.168.255.3

OS and Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 16.19 seconds

┌──(kali㉿kali)-[~]
└─$
```

Aggressive scan against .3.

8. Explain what the -p1-1024 switch does.

The -p1-1024 switch restricts the port scan to a specific range — in this case, ports 1 through 1024, the traditional range of "well-known"/system ports (e.g., FTP 21, SSH 22, Telnet 23, HTTP 80). By default, Nmap scans its top 1000 most common ports across the full 0–65535 range; specifying -p1-1024 narrows and speeds up the scan when only classic, privileged system services are of interest, rather than the full port space.

Scan 16 of 17 — I ran this from a terminal on my Kali machine:

```
sudo nmap -Pn 192.168.255.2 -p1-1024
```

```
┌──(kali㉿kali)-[~]
└─$ sudo nmap -Pn 192.168.255.2 -p1-1024
Starting Nmap 7.99 ( https://nmap.org ) at 2026-09-26 13:57 -0400
Nmap scan report for 192.168.255.2
Host is up (0.00052s latency).
Not shown: 1018 closed tcp ports (reset)
PORT    STATE SERVICE
21/tcp  open  ftp
22/tcp  open  ssh
80/tcp  open  http
135/tcp open  msrpc
139/tcp open  netbios-ssn
445/tcp open  microsoft-ds
MAC Address: 08:00:27:D7:CC:D8 (Oracle VirtualBox virtual NIC)

Nmap done: 1 IP address (1 host up) scanned in 1.92 seconds
```

Well-known ports 1–1024 against .2, with host discovery skipped.

Scan 17 of 17 — I ran this from a terminal on my Kali machine:

```
sudo nmap -Pn 192.168.255.3 -p1-1024
```

```
┌──(kali㉿kali)-[~]
└─$ sudo nmap -Pn 192.168.255.3 -p1-1024
Starting Nmap 7.99 ( https://nmap.org ) at 2026-09-26 13:41 -0400
Nmap scan report for 192.168.255.3
Host is up (0.00093s latency).
Not shown: 1017 closed tcp ports (reset)
PORT    STATE SERVICE
21/tcp  open  ftp
22/tcp  open  ssh
80/tcp  open  http
111/tcp open  rpcbind
139/tcp open  netbios-ssn
445/tcp open  microsoft-ds
631/tcp open  ipp
MAC Address: 08:00:27:42:51:79 (Oracle VirtualBox virtual NIC)

Nmap done: 1 IP address (1 host up) scanned in 1.02 seconds

┌──(kali㉿kali)-[~]
└─$
```

Well-known ports 1–1024 against .3.

Summary

The experiment showed how much the selection of an Nmap option affects both the information obtained about the target and how noisy or detectable the scan is. The first -sn scan discovered all three hosts on the lab network, and each of -sS, -sT and -sU verified the same basic set of open ports on .2 and .3 by means that were different — -sS and -sT agreed very closely on the TCP side, while -sU revealed a completely different set of UDP services (such as SNMP and NetBIOS) which the TCP-only scans could not have detected at all.

The -O scan properly identified the two machines as running different operating systems without needing to log in to either of them: machine .2 was identified as Windows Server 2008 R2 (with estimates spanning Vista/7/8.1) and machine .3 as having a Linux kernel version 3.x or 4.x. The -sV option then provided the exact versions on that basis—OpenSSH 6.6.1p1 and Apache 2.4.7 on the Ubuntu system, IIS httpd 7.5 and MySQL 5.5.20 on the Windows system—details of the sort that would be important when identifying known vulnerabilities.

The -A scans put all of this together and also brought to light the lab's most interesting discovery: on .3, the Apache http-ls script showed the contents of the web root directory, including a phpmyadmin/ folder and a file called payroll_app.php which was located in a directory that was publicly browsable. In neither case was it necessary to use any credentials or carry out active exploitation in order to discover this — rather, the information was obtained through normal service enumeration, which is precisely the aim of this exercise, since a great deal of useful information about a target can be gathered before an attacker even attempts to break in.